

Mathematics, Computation, and Linguistic Structure

What descriptions, mechanisms, theories, and models each claim

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A formal description is not automatically a mechanism

Question

What does a mathematical success establish about linguistic reality?

- A description can characterize a pattern without computing it.
- A computation requires states, transitions, input, and an acceptance condition.
- A model connects selected structure to an empirical target.
- A theory explains why the connection should hold.

MOLR §§1.8–1.12

Where we are

1. Mathematics and computation
2. Linguistic structure
3. Philosophy and typology
4. Theories and models

Abstract computation suppresses its physical encoding

Abstract machine

States and transitions define which formal configurations follow which.

Physical computer

Encoding and decoding connect material states to those formal configurations.

Take-away

A mathematical mapping does not identify the physical system that realizes it.

MOLR §1.8

Descriptions answer different questions

Description	Which objects satisfy a condition?
Grammar	Which strings or structures are admitted?
Algorithm	Which effective steps produce an answer?
Machine	Which configurations realize those steps?
Model	Which formal relations correspond to an empirical target?

Guided analysis

Classify a parse tree, a CKY procedure, and a corpus frequency table.

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Linear order does not determine dependency

The student that I persuaded / promised / expected / wanted to study arrived.

- The same surface position participates in different lexical dependencies.
- The understood subject of *to study* changes with the embedding verb.
- A string alone underdetermines the structure relevant to interpretation.

Guided analysis

For each verb, identify who is understood to study.

MOLR §1.9

Grammaticality, meaningfulness, and truth can diverge

- ① Colorless green ideas sleep furiously.
 - ② Green furiously ideas sleep colorless.
- Formal well-formedness is not identical to ordinary interpretability.
 - Interpretability is not identical to factual truth.
 - Judgments can vary with context, construction, and speaker experience.

Question

What exactly is marked by the star in linguistic practice?

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Grammar is an INUS condition

INUS: an insufficient but necessary part of an unnecessary but sufficient condition.

For a successful linguistic act

Grammar may be necessary within one sufficient package, while context, shared knowledge, attention, and social conditions are also required.

Take-away

Calling grammar necessary need not make it sufficient, exclusive, or universal in its particular form.

MOLR §1.10

Typology compares variants through explicit invariants

- A typology requires a comparison class.
- A category must be defined independently of any one language's exponent.
- Cross-linguistic similarity may concern dependency, function, position, or interpretation.
- Treating a familiar form as the invariant builds the conclusion into the method.

Guided analysis

How could “relative clause” be defined without requiring an English-like relative pronoun?

MOLR §1.11

Relative clauses preserve a relation across formal variation

$$rel(P, Q) = \lambda x. P(x) \wedge Q(x)$$

- Languages may use gaps, resumptive elements, nominalization, or other devices.
- The semantic intersection is one candidate invariant.
- The syntactic means remain empirical variants, not defective realizations of one form.

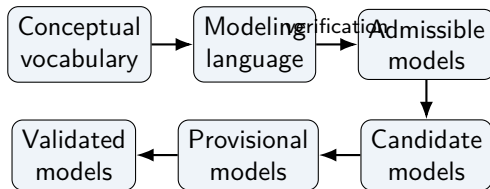
Question

Which facts would show that this semantic abstraction is too weak?

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A model is selected through several admissibility filters



Take-away

Formal admissibility narrows a space; empirical validation decides whether the surviving model earns its interpretation.

MOLR §1.12

Verification and validation are different achievements

Verification

Does the formal construction satisfy the rules of the modeling language?

Validation

Does the construction correspond adequately to the linguistic phenomenon?

Guided analysis

A grammar generates exactly the corpus sentences. Is it verified? Is it validated? What further evidence is needed?

Third hour: untangle the claims

For each statement, identify whether it concerns description, computation, modeling, or explanation.

- 1 “This grammar generates $a^n b^n$.”
- 2 “This parser terminates for every input.”
- 3 “This dependency corresponds to speaker judgments.”
- 4 “Children converge because this constraint is preserved.”

Guided analysis

Give one way in which each claim could be correct while the next stronger claim is false.

Week 3 synthesis

- ① Mathematical description and computation are related but distinct.
- ② Linguistic structure cannot be read directly from linear order.
- ③ Grammar participates in sufficient packages without exhausting them.
- ④ Typology requires explicit invariants across genuine variants.
- ⑤ Models must be formally verified and empirically validated.

Question

Quiz 2 tests whether you can keep these kinds of claim apart.